

Fundamentals of API Development

Optional Breakout: Building API Endpoints

Day 3 - 30-minute add-on

Lab Overview

Goal: Use a supplied prompt and acceptance tests to add one API challenge to your local quickstart project.

- **Format:** Teams choose one of six challenges
- **Basic:** Controller-only endpoint
- **Intermediate:** Controller delegates to a service
- **Stretch:** Multi-operation API with records, state, or validation

This is an optional add-on and is not included in Day 3's 240-minute schedule.

Setup and Workflow

1. Open `api-dev-setup/quickstart-project/breakouts/build-api-endpoints/`
2. Choose one challenge appropriate for your team
3. Copy its `start/` files into the matching quickstart `src/` paths
4. Copy the challenge prompt into your coding assistant and review its proposed changes
5. Run the supplied tests, then compare with `solution/` only if needed

Choose Your Challenge

One team, one endpoint set, one passing test suite

Basic: Controller Only

Challenge	Endpoint	Result
Square a number	<code>GET /api/v1/math/square?number=7</code>	Return the input and its square
Describe text	<code>GET /api/v1/text/{word}/details</code>	Return the original word, length, and uppercase form

Choose basic if your team wants to practice request mapping, parameters, path variables, and JSON responses.

Intermediate: Controller and Service

Challenge	Endpoint	Responsibility
Shipping estimate	GET /api/v1/shipping/estimate	Controller handles HTTP; service calculates deterministic pricing
Tax calculation	GET /api/v1/tax/calculate	Controller handles inputs; service applies a regional rate

Use constructor injection. Keep calculation logic out of the controller.

Stretch: More Complex APIs

Challenge	Operations	Added design decision
Inventory	List, find by SKU, create	In-memory state plus <code>201</code> , <code>404</code> , and conflict handling
Order pricing	Price a multi-line order	Request records, totals, discount, tax, and invalid-request handling

These are stretch options, not prerequisites for Day 4.

Use the Prompt, Then Apply Judgment

1. Give the coding assistant the supplied prompt and the starter files
2. Require it to preserve package names, paths, dependencies, and test expectations
3. Review controller responsibilities, service boundaries, status codes, and response fields
4. Run the focused tests instead of accepting the answer by inspection
5. Ask for a correction using the exact failure output when a test fails

Completion Checklist

- ☐ One challenge selected and its starter files copied into the quickstart project
- ☐ Supplied prompt used without changing the acceptance criteria
- ☐ Generated code reviewed before running it
- ☐ Challenge tests passing with `mvn test`
- ☐ Endpoint exercised once through curl, Postman, or Swagger UI

Share Your Result

Endpoint, design choice, and one correction you made to the generated code